Intra-Cell Spatial Reuse in LEO Downlinks: Beam-Leakage-Aware Satellite Scheduling

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***Abstract*—Dense low-earth-orbit (LEO) constellations offer an opportunity to increase local broadband capacity by serving multiple terminals in the same cell through concurrent satellite beams. However, interference between overlapping beams limits spatial reuse, and satellite selection based solely on link strength or angular separation may result in suboptimal performance. This paper proposes beam-leakage-aware satellite scheduling for Ku-band LEO downlink transmissions. With matched transmit and receive beamforming, we formulate a fixed-cardinality sum-rate maximization problem that jointly accounts for satellite-dependent received powers and pairwise receive-array leakage. A beam-leakage-aware branch-and-bound (BL-BnB) algorithm combines an admissible upper bound with greedy-plus-swap initialization to obtain globally optimal schedules for the reference-point model. We evaluate the method using a 24-h ephemeris-derived Starlink visibility trace with antenna and transmit-power models calibrated to public SpaceX filings. The results show that total capacity increases with beam count, but less than proportionally. With eight concurrent beams, BL-BnB improves mean per-beam capacity by 27.2% over nearest-satellite selection, primarily benefiting weaker links. BL-BnB achieves millisecond-scale runtimes in our evaluation, supporting the computational feasibility of optimal scheduling.**

***Index Terms*—LEO satellites, spatial reuse, satellite scheduling, interference management, branch-and-bound.**

I. Introduction

Local traffic demand in low-Earth-orbit (LEO) broadband networks can exceed the capacity of a single satellite beam, motivating concurrent service from multiple simultaneously visible satellites [1]. In a recreational vehicle (RV) park or at rural terminal densities, active terminals can be too close together to be served by non-overlapping spot beams. Concurrent beams from multiple satellites can increase local capacity by reusing the same time–frequency resource. The resulting intra-cell spatial reuse, illustrated in Fig. 1, depends on how effectively the terminals can separate desired and interfering downlinks.

Regulatory developments further motivate the study of overlapping beams. To protect geostationary-satellite-orbit (GSO) systems under the International Telecommunication Union (ITU)'s interference limits [2], the Federal Communications Commission (FCC) initially restricted Starlink Gen2 to one co-frequency, co-polarization beam per location in the 12.2–12.7 and 19.7–20.2 GHz bands [3]. Then, its January 2026 waiver permitted up to eight overlapping Ku-band beams with 4° GSO-arc avoidance [4]. The framework adopted in April 2026 introduced an alternative that does not explicitly limit the number of overlapping beams but instead relies on GSO-link performance protection and 3° arc avoidance in the United States [5]. These developments make the capacity benefits and interference costs of concurrent beams increasingly relevant.

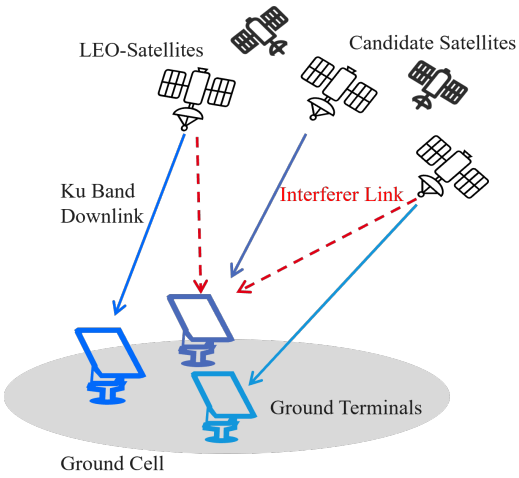


Fig. 1: Considered Ku-band LEO downlink with multiple ground terminals and candidate satellites. Reuse of the same time–frequency resource creates inter-beam interference between concurrently active links.

The most critical challenge is that satellite visibility does not directly translate into usable spatial reuse. For a compact cell, satellite transmit footprints substantially overlap, so interference rejection relies primarily on the ground terminals' receive arrays. Selecting the nearest satellites favors short propagation paths but can also end up with highly correlated arrival directions. Conversely, enforcing a large angular separation can be unnecessarily restrictive, sacrificing desired-link strength without minimizing the actual array response to interference. Since each active satellite contributes interference to all other scheduled links, the achievable sum rate depends jointly on individual received powers and aggregate beam leakage.

Prior work has explored spatial processing and resource allocation to control satellite interference. Massive multiple-input multiple-output (MIMO) transmission combines statistical channel information and space-angle-based user grouping for full-frequency reuse, but considers a single satellite serving all scheduled users [6]. Cooperative downlinks jointly optimize hybrid beamforming and user scheduling, assum-

ing inter-satellite data exchange [7]. Game-theoretic transmit covariance optimization mitigates uplink jamming, assuming cooperative satellite reception [8]. Interference prediction and hybrid receive beamforming improve long-term rates through receiver adaptation rather than satellite subset selection [9]. Coordinated beam hopping balances loads using spatially isolated transmission patterns [10], restricting simultaneous illumination of the same cell. These approaches leave open the achievable intra-cell reuse from satellite selection alone with fixed matched beamformers.

More closely related, ephemeris-based min–max satellite selection (E-MMS) greedily minimizes the largest receive-array correlation with already selected satellites [1]. This criterion measures spatial separability. However, the desired received power (the beamformed signal power from the intended satellite) varies across candidate links with propagation distance, antenna gain, and transmit power. Each link's rate also depends on the sum of interference from all other selected satellites, which the largest pairwise correlation alone does not capture. Thus, correlation-based selection does not guarantee the maximum sum rate. Computing the globally optimal sum rate under the same model provides a reference for assessing heuristic selection.

This paper studies concurrent Ku-band LEO downlinks with matched transmit and receive beamforming. We select a prescribed number of satellites at each scheduling interval using a common cell reference point. A gateway can estimate the scheduling inputs from satellite ephemerides, terminal Global Positioning System (GPS) coordinates, known beam powers and array responses, and precomputed loss tables based on ITU Radiocommunication Sector (ITU-R) Recommendation P.618 [11]. Our contributions are threefold:

- **Beam-leakage-aware scheduling formulation:** We formulate fixed-cardinality intra-cell sum-rate maximization using satellite-dependent received powers and pairwise receive-array leakage under full time–frequency reuse.
- **Globally optimal satellite selection:** Beam-leakage-aware branch-and-bound (BL-BnB) combines an admissible upper bound with greedy-plus-swap initialization to certify optimality for the reference-point model. Selecting eight of 28 candidates is approximately 267× faster than exhaustive search on average.
- **Practical spatial-reuse assessment:** A 24-h Starlink trace and models calibrated to public SpaceX filings quantify the effects of beam count, cell radius, and scheduling. Mean per-beam capacity improves over nearest-satellite selection by 27.2% with eight beams and 9.8% with 49 beams, primarily benefiting weaker links.

II. Problem Formulation

We consider an LEO downlink where multiple visible satellites serve a ground cell on a shared time–frequency resource. In each scheduling interval, geometry and large-scale channel states are fixed, and K concurrent layers are selected to maximize the instantaneous cell sum rate, accounting for link strength and inter-layer beam leakage.

A. System Model and Assumptions

At time instant t, let $\mathcal{S}(t)$ denote the set of satellites that satisfy the visibility and link-availability requirements, and denote $N(t) = |\mathcal{S}(t)|$ as the number of satellites in this set. We focus on intervals with $N(t) \geq K$, and if fewer than K satellites are visible, the scheduler activates all available satellites and replaces K by $N(t)$. Each selected satellite provides one downlink layer to the cell. Because the cell diameter is much smaller than the LEO slant range, the scheduling metrics are evaluated at a common cell reference point $\mathbf{r}_\mathrm{c}$, assuming identical receive arrays with the same physical orientation. Each satellite carries independent data for one terminal and satellite–user assignment is fixed.

For satellite s at point $\mathbf{r_s}$, the slant range is $d_s(t) = \|\mathbf{r}_s(t) - \mathbf{r}_\mathrm{c}\|_2$, giving free-space loss $\beta_s^{\mathrm{FSPL}}(t) = (\lambda/(4\pi d_s(t)))^2$ at wavelength λ. The effective large-scale gain is

$$\beta_s^{\mathrm{eff}}(t) = \beta_s^{\mathrm{FSPL}}(t) A_s^{\mathrm{atm}}(t) A_s^{\mathrm{rain}}(t) A_s^{\mathrm{pol}}(t) A_s^{\mathrm{misc}}(t), \tag{1}$$

where the factors account for non-rain atmospheric, rain, polarization, and other large-scale losses.

Each satellite employs an N_t-element transmit array, and each ground receiver employs an N_r-element receive array. Under the dominant LoS assumption, the channel associated with satellite s is

$$\mathbf{H}_s(t) = \sqrt{\beta_s^{\mathrm{eff}}(t) G_{\mathrm{t},s}(t) G_{\mathrm{r},s}(t)}\, \mathbf{a}_{\mathrm{r},s}(t) \mathbf{a}_{\mathrm{t},s}^{\mathrm{H}}(t), \tag{2}$$

where $G_{\mathrm{t},s}$ and $G_{\mathrm{r},s}$ are linear element-pattern gains in the departure and arrival directions, respectively. A common link phase is omitted because it cancels in the received-power expressions. The transmit and receive steering vectors, i.e., $\mathbf{a}_{\mathrm{t},s}(t)$ and $\mathbf{a}_{\mathrm{r},s}(t)$, have squared norms N_t and N_r, respectively. For the rank-one channel in (2), dominant-mode singular value decomposition (SVD) beamforming is equivalent to matched steering, $\mathbf{f}_s = \mathbf{a}_{\mathrm{t},s}/\sqrt{N_\mathrm{t}}$ and $\mathbf{w}_s = \mathbf{a}_{\mathrm{r},s}/\sqrt{N_\mathrm{r}}$. With transmit power $P_{\mathrm{tx},s}$, the desired post-beamforming received power is therefore

$$q_s(t) = P_{\mathrm{tx},s} \beta_s^{\mathrm{eff}}(t) G_{\mathrm{t},s}(t) G_{\mathrm{r},s}(t) N_\mathrm{t} N_\mathrm{r}. \tag{3}$$

B. Beamformed Signal and Interference Model

When the receiver is steered toward satellite s, a cochannel signal from satellite s' is attenuated according to the receive-array de-correlation. We define the normalized leakage coefficient as

$$\eta_{s,s'}(t) = \frac{\left|\mathbf{a}_{\mathrm{r},s}^{\mathrm{H}}(t) \mathbf{a}_{\mathrm{r},s'}(t)\right|^2}{N_\mathrm{r}^2}, \qquad 0 \leq \eta_{s,s'}(t) \leq 1. \tag{4}$$

Consequently, the interference power from satellite s' to the layer associated with satellite s is $q_{s'}(t)\eta_{s,s'}(t)$.

For a scheduled subset $\mathcal{A}(t) \subseteq \mathcal{S}(t)$ with $|\mathcal{A}(t)| = K$, the signal-to-interference-plus-noise ratio (SINR) of layer s is

$$\gamma_s(\mathcal{A}, t) = \frac{q_s(t)}{\sigma^2 + \sum\limits_{\substack{s' \in \mathcal{A}(t) \\ s' \neq s}} q_{s'}(t) \eta_{s,s'}(t)}. \tag{5}$$

Algorithm 1 BL-BnB Satellite Scheduling

Input: $\mathcal{S}$, $\{q_s\}$, $\{\eta_{s,j}\}$, K, B, and σ^2
Output: Globally optimal set $\mathcal{A}^\star$ and sum rate $C^\star$

Set $g_{s,j} \leftarrow q_j \eta_{s,j}$ and $g_{s,s} \leftarrow 0$
Initialize $\mathcal{A}^\star$ by maximizing C_{sum} over the top-K and bottom-K power sets and the greedy–swap set
$C^\star \leftarrow C_{\text{sum}}(\mathcal{A}^\star)$ and $\mathcal{Q} \leftarrow [(\emptyset, \mathcal{S})]$
while $\mathcal{Q} \neq \emptyset$ **do**
  Pop $(\mathcal{P}, \mathcal{R})$ from $\mathcal{Q}$ and set $m \leftarrow K - |\mathcal{P}|$
  if $m < 0$ or $m > |\mathcal{R}|$ **then**
    continue
  else if $m = 0$ **then**
    $\mathcal{A} \leftarrow \mathcal{P}$
  else if $m = |\mathcal{R}|$ **then**
    $\mathcal{A} \leftarrow \mathcal{P} \cup \mathcal{R}$
  else
    Compute $\text{UB}(\mathcal{P}, \mathcal{R})$ using (10)
    if $\text{UB}(\mathcal{P}, \mathcal{R}) \le C^\star$ **then**
      continue
    end if
    $c^\star \leftarrow \arg\max_{c \in \mathcal{R}} \bar{R}_c^{(\mathcal{P},\mathcal{R})}$ from (9)
    $\mathcal{R}' \leftarrow \mathcal{R} \setminus \{c^\star\}$
    Push $(\mathcal{P}, \mathcal{R}')$ and then $(\mathcal{P} \cup \{c^\star\}, \mathcal{R}')$ onto $\mathcal{Q}$
    continue
  end if
  $C \leftarrow C_{\text{sum}}(\mathcal{A})$
  If $C > C^\star$, set $(\mathcal{A}^\star, C^\star) \leftarrow (\mathcal{A}, C)$
end while
return $\mathcal{A}^\star$ and $C^\star$

Here, $\sigma^2 = N_0 B$, where N_0 is the effective noise power spectral density, including the receiver noise figure, and B is the bandwidth. Treating inter-layer interference as noise, the per-layer rate is $R_s(\mathcal{A}, t) = B \log_2(1 + \gamma_s(\mathcal{A}, t))$, and the cell sum rate is $C_{\text{sum}}(\mathcal{A}, t) = \sum_{s \in \mathcal{A}(t)} R_s(\mathcal{A}, t)$.

C. Scheduling Objective

The scheduler selects exactly K satellites to maximize the cell sum rate. Let $y_s(t)$ indicate whether satellite s is scheduled, so that $\mathcal{A}(t) = \{s \in \mathcal{S}(t) : y_s(t) = 1\}$. Using the SINR in (5), we solve

$$\mathbf{y}^\star(t) \in \arg\max_{\mathbf{y}(t)} \sum_{s \in \mathcal{S}(t)} y_s(t) B \log_2\big(1 + \gamma_s(\mathcal{A}, t)\big) \tag{6a}$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}(t)} y_s(t) = K, \quad y_s(t) \in \{0, 1\}, \ \forall s \in \mathcal{S}(t). \tag{6b}$$

The objective is nonseparable because each selected satellite changes the interference at the other scheduled layers. Maximizing the sum rate therefore requires jointly accounting for the received powers $\{q_s(t)\}$ and pairwise leakage coefficients $\{\eta_{s,s'}(t)\}$.

III. Optimal Scheduling

Problem (6) admits $\binom{N(t)}{K}$ candidate subsets. We obtain the globally optimal schedule using BL-BnB (Algorithm 1). It maintains the best feasible schedule found so far (the incumbent) and prunes branches whose admissible upper bounds do not exceed its sum rate.

A. Beam-Leakage-Aware Upper Bound

The time index is omitted in this section. A search node $(\mathcal{P}, \mathcal{R})$ consists of a partial selection $\mathcal{P}$ and a disjoint pool of undecided candidates $\mathcal{R}$. With $m = K - |\mathcal{P}|$, the node is infeasible if $m < 0$ or $m > |\mathcal{R}|$. Each choice of $\mathcal{T} \subseteq \mathcal{R}$ with $|\mathcal{T}| = m$ completes the schedule as $\mathcal{A} = \mathcal{P} \cup \mathcal{T}$, where $|\mathcal{A}| = K$.

For a satellite s and candidate set $\mathcal{U}$, define the minimum additional interference contributed by any m members of $\mathcal{U}$ as

$$L_s(\mathcal{U}, m) = \min_{\substack{\mathcal{B} \subseteq \mathcal{U} \\ |\mathcal{B}| = m}} \sum_{j \in \mathcal{B}} q_j \eta_{s,j}, \qquad L_s(\mathcal{U}, 0) = 0. \tag{7}$$

Thus, $L_s(\mathcal{U}, m)$ is the sum of the m smallest values in $\{q_j \eta_{s,j} : j \in \mathcal{U}\}$.

For an already selected satellite $s \in \mathcal{P}$, define an upper bound on its rate in any feasible completion as

$$\bar{R}_s^{(\mathcal{P},\mathcal{R})} = B \log_2\left(1 + \frac{q_s}{\sigma^2 + \sum\limits_{\substack{j \in \mathcal{P} \\ j \neq s}} q_j \eta_{s,j} + L_s(\mathcal{R}, m)}\right). \tag{8}$$

For $m \ge 1$ and an undecided satellite $c \in \mathcal{R}$, selection of c requires $m - 1$ additional satellites from $\mathcal{R} \setminus \{c\}$, giving

$$\bar{R}_c^{(\mathcal{P},\mathcal{R})} = B \log_2\left(1 + \frac{q_c}{\sigma^2 + \sum\limits_{j \in \mathcal{P}} q_j \eta_{c,j} + L_c(\mathcal{R} \setminus \{c\}, m - 1)}\right). \tag{9}$$

Let $\text{Top}_m(\mathcal{R})$ contain the m undecided satellites with the largest values of (9). An upper bound on the best sum rate reachable from node $(\mathcal{P}, \mathcal{R})$ with $m \ge 1$ is

$$\text{UB}(\mathcal{P}, \mathcal{R}) = \sum_{s \in \mathcal{P}} \bar{R}_s^{(\mathcal{P},\mathcal{R})} + \sum_{c \in \text{Top}_m(\mathcal{R})} \bar{R}_c^{(\mathcal{P},\mathcal{R})} \tag{10}$$

Theorem 1. *For every feasible completion $\mathcal{T} \subseteq \mathcal{R}$ of a nonterminal node with $|\mathcal{T}| = m \ge 1$,*

$$C_{\text{sum}}(\mathcal{P} \cup \mathcal{T}) \le \text{UB}(\mathcal{P}, \mathcal{R}) \tag{11}$$

Proof: For each $s \in \mathcal{P}$, the interference contributed by any feasible completion $\mathcal{T}$ is no smaller than $L_s(\mathcal{R}, m)$. Similarly, for each $c \in \mathcal{T}$, the interference from the other $m-1$ newly selected satellites is no smaller than $L_c(\mathcal{R} \setminus \{c\}, m-1)$. Because the rate is monotonically decreasing in interference, the corresponding rates are bounded by (8) and (9). Finally, any completion contains exactly m candidates, whose total contribution cannot exceed the sum of the m largest candidate bounds in (10).

B. Branch-and-Bound Search and Initialization

The search starts at $(\emptyset, \mathcal{S})$. A node is infeasible if $m < 0$ or $m > |\mathcal{R}|$, meaning that too many satellites have been selected or too few candidates remain to reach K. Such nodes are discarded. If $m = 0$, the only completion is $\mathcal{P}$; if $m = |\mathcal{R}|$, it is $\mathcal{P} \cup \mathcal{R}$. In either case, its sum rate is evaluated using the original objective, and the incumbent is updated if improved.

For a nonterminal node, pruning occurs when $\text{UB}(\mathcal{P}, \mathcal{R}) \le C^\star$, where $C^\star$ is the best feasible sum rate found so far. By the preceding theorem, no completion of this node can improve

$C^\star$. Otherwise, the candidate $c^\star$ specified in Algorithm 1 generates two children: $(\mathcal{P}\cup\{c^\star\}, \mathcal{R}\setminus\{c^\star\})$ and $(\mathcal{P}, \mathcal{R}\setminus\{c^\star\})$. The inclusion child is explored first to seek early incumbent improvements. Once the search is exhausted, the incumbent is globally optimal.

Initialization greedily adds the candidate yielding the largest resulting sum rate until K satellites are selected. One-swap refinement then repeatedly accepts the best strictly improving exchange between a selected and an unselected satellite until none remain. The best of this set and the top-K and bottom-K received-power sets initialize the incumbent. This affects search efficiency, not optimality. Algorithm 1 specifies the branching rule and complete procedure.

C. Complexity Analysis

Precompute $g_{s,s'} = q_{s'}\eta_{s,s'}$. In this way, evaluating one subset of size K requires $O\left(K^2\right)$ operations, giving exhaustive cost $O\left(\binom{N}{K}K^2\right)$ and interference-matrix storage $O\left(N^2\right)$.

Computing all pairwise receive-array correlations by explicit inner products of length N_r requires $O\left(N^2N_\mathrm{r}\right)$ operations. At each BL-BnB node, the implemented upper-bound calculation fully sorts up to N interference lists, each containing at most N values. Its worst-case cost is therefore $O\left(N^2\log N\right)$. If M_BB nodes are visited and I_sw swap iterations are performed, the overall complexity is

$$O\Big(N^2N_\mathrm{r} + (1+I_\mathrm{sw})NK^3 + M_\mathrm{BB}N^2\log N\Big) \tag{12}$$

BL-BnB remains exponential in the worst case as M_BB may approach exhaustive enumeration, while admissible pruning reduces practical search cost while preserving global optimality.

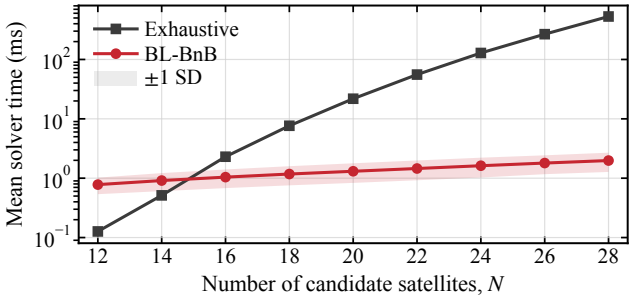


Fig. 2: Solver runtime (ms) in log-scale for selecting $K = 8$ of N candidates. Curves show the mean of 720 per-snapshot mean runtimes (three repeats per method and snapshot). Shading indicates ±1 sample standard deviation across these means.

Fig. 2 compares BL-BnB and vectorized exhaustive search on identical strongest-N sets ($K = 8$), timing only initialization and search. Initialization and bounding overhead make BL-BnB slower at small N, but it is faster at every tested $N \geq 16$. At $N = 28$, mean runtime falls from 529.48 to 1.98 ms (about 267×). With all 49–90 visible candidates, BL-BnB takes 4.83 ms (median of 720 per-snapshot medians, each from three repeats).

TABLE I: Simulation Parameters.

Radio Parameters	**Values**
Carrier/Bandwidth	11.7 GHz/250 MHz
Noise PSD/Noise Figure	−174 dBm/Hz/5 dB
TX/RX Arrays	30 × 30/20 × 23 uniform rectangular arrays (URAs)
Channel/Beamforming	Matched steering, ITU-R P.618 losses [11]
Scene Configuration	
Reference Location	39.95697°N, 105.16033°W
Duration/Sampling	24 h, 720 snapshots, 2-min interval
Orbit/Elevation Mask	Gen-1/Gen-2 LEO, 25°
Cell Radius	$R \in \{0.25, 5, 15\}$ km
Paired Beams	$K \in \{1, 4, 8, 14, 20, 30, 40, 49\}$
Spectrum Reuse	Full time–frequency reuse

IV. Practical Study and Performance Analysis

A. Simulation Methodology

We evaluate 720 snapshots sampled at 2-min intervals over a 24-h Starlink trace generated from CelesTrak orbital elements [12] (Table I), with 49–90 visible satellites per snapshot. Both methods select K visible satellites, with one beam per satellite serving an independently generated user on the shared time–frequency resource. The baseline chooses the K nearest satellites, whereas BL-BnB maximizes the reference-point sum rate. Using the same SINR model, we apply the 3GPP implementation-loss factor of 0.6 [13] and set the SINR cap to 30 dB. The objective, initialization, bounds, and evaluation use the resulting nondecreasing mapping $r(\gamma) = 0.6B\log_2\left(1+\min\left\{\gamma, 10^3\right\}\right)$. Evaluation uses user locations. Unless noted, $R = 250$ m. The rates provide a snapshot-level benchmark for spatial reuse. Handover and other unmodeled overhead would reduce net throughput. For each elevation angle, ITU-R P.618 provides the attenuation level exceeded for 1% of an average year. These values are precomputed on a 0.1° elevation grid.

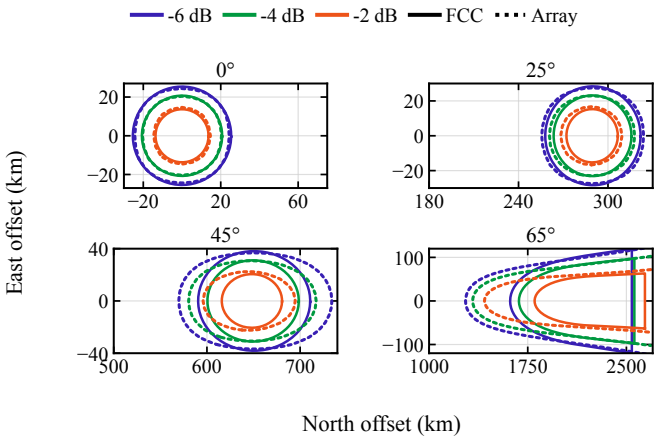


Fig. 3: Ground gain contours of SpaceX Gen2 [14] (solid) and the modeled 30×30 array (dotted) at 11.7 GHz and 614-km altitude. Scan angles: 0°, 25°, 45°, 65°. Contour levels: −6, −4, −2 dB (colored).

We model and calibrate the 30 × 30 array to SpaceX Gen2 low-gain Ku-band beams [14], adjusting element gain for 34-dBi boresight gain under the conditions in Fig. 3. Modeled contours closely match the filing at 0° and 25° but extend

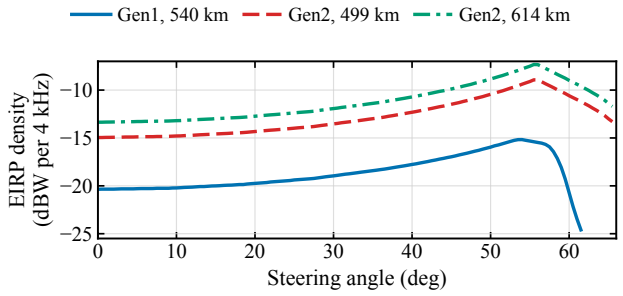


Fig. 4: Starlink EIRP density versus steering angle for Gen-1 at 540 km [15] and Gen-2 at 499 km (approximated) and 614 km [14], in dBW per 4 kHz.

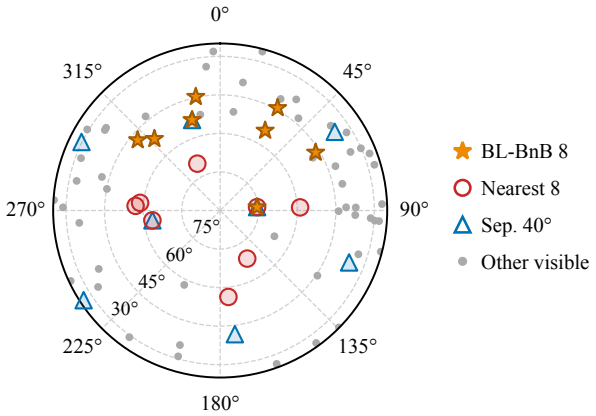


Fig. 5: Satellite-selection sky view: overlaid markers indicate shared satellites. Gray dots denote other visible satellites. Outer labels and rings indicate azimuth and elevation, respectively.

beyond it at 45° and 65°. Although larger arrays could narrow these footprints, we retain the fixed size because the studied cells ($R \leq 15$ km) remain within the filed −2-dB contours at both angles. For the RV-park terminals, the 20×23 array approximates Starlink Mini's 298.5×259-mm dimensions [16] and matches the 30.4-dBi receive gain of SpaceX's ESIM3-V2 terminal [17].

Transmit power follows the digitized steering-dependent equivalent isotropically radiated power (EIRP) curves [18] in Fig. 4. The 499-km curve shifts the 614-km curve down by 1.6 dB to match the filed maximum EIRP density.

B. Intra-cell Performance

In Fig. 6, at $R = 250$ m and $K = 1, 8, 49$, mean per-beam capacities are 451, 363, and 204 Mbps, while mean totals are 0.45, 2.90, and 9.98 Gbps, respectively. Thus, 49 overlapping beams provide only 22.2× the single-beam capacity, with declining per-beam rates. Additional beams reuse the full bandwidth but increase aggregate receive-array leakage, limiting spatial multiplexing gains.

Increasing R to 5 km changes the mean by less than 0.3% for every K. As the insets show, differences emerge mainly at 15 km with many beams, where the $K = 49$ mean total rises from 9.98 to 10.21 Gbps (2.3%). The weak radius dependence is consistent with modest variations in slant range and arrival geometry. With substantially overlapping satellite main lobes, interference rejection relies primarily on receive beamforming to resolve arrival directions.

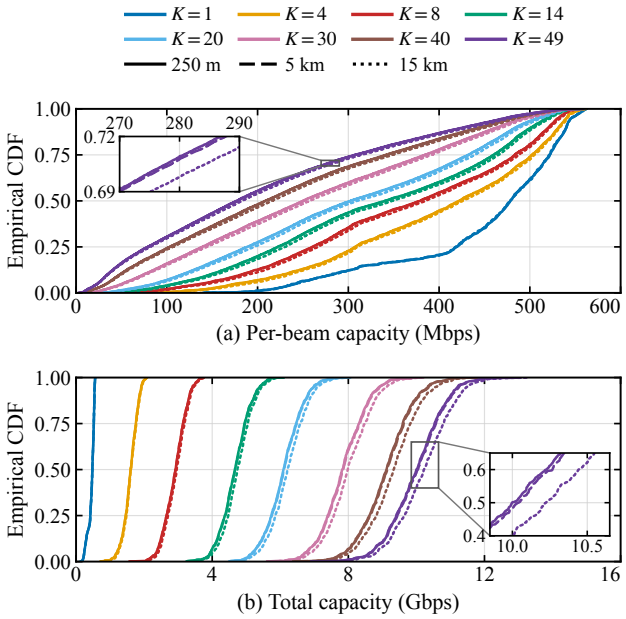


Fig. 6: Empirical CDFs of (a) per-beam and (b) total capacity. Colors denote the number of paired beams K. Line styles denote cell radii of 0.25, 5, and 15 km. Insets magnify the boxed regions for $K = 49$.

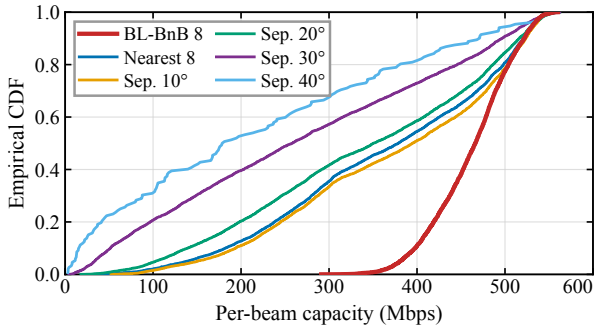


Fig. 7: Per-beam capacity CDFs for eight paired beams under BL-BnB, nearest-satellite selection, and spherical-separation baselines. The 40° curve uses only the 19 snapshots where eight satellites can be selected.

C. Optimal Scheduling Performance

The eight-beam policy CDF uses matched inputs. Beam-count comparisons include unpaired archived user scenes at $K = 8, 14, 30$ and slightly different propagation losses.

Nearest-eight selection favors high elevations, potentially clustering arrival directions and increasing receive-array correlation (Fig. 5). In Fig. 7, a 10° minimum spherical separation raises mean per-beam capacity from 362.9 to 372.9 Mbps (2.8%), whereas 20° and 30° reduce it to 339.8 and 267.7 Mbps. Thus, wider separation alone does not optimize the signal–interference tradeoff. E-MMS [1] instead greedily minimizes maximum pairwise receive-array correlation, differing from our fixed-angle baseline without guaranteeing sum-rate optimality. BL-BnB optimizes reference-point sum capacity globally, averaging 461.6 Mbps per beam (27.2% above nearest-eight selection).

In Fig. 8(a), mean per-beam gain falls from 98.7 Mbps at $K = 8$ to 20.0 Mbps at $K = 49$. Relative gains are 27.2%,

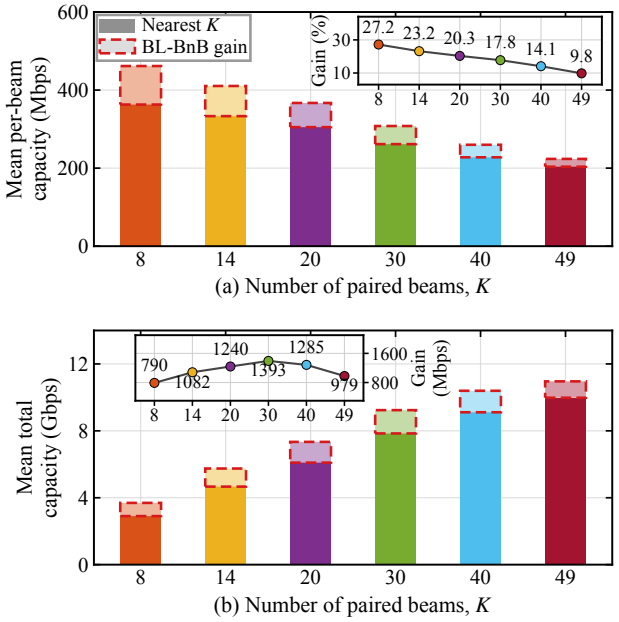


Fig. 8: Mean (a) per-beam and (b) total capacity versus K. Colored bars show nearest-K selection. Red dashed extensions show BL-BnB gains. Insets show (a) percentage and (b) absolute total-capacity gains over nearest-K selection.

17.8%, and 9.8% at $K = 8, 30, 49$, respectively. Activating more visible satellites leaves fewer choices to avoid weak links or strong interference. Total gain equals K times the mean per-beam gain: 790, 1393, and 979 Mbps at $K = 8, 30, 49$, respectively (see Fig. 8(b)). Beyond $K = 30$, declining per-beam gains outweigh the increasing beam count.

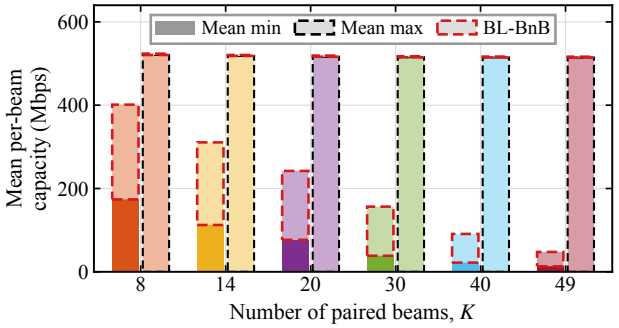


Fig. 9: Time-averaged per-snapshot minimum (solid) and maximum (black dashed) beam capacities versus K for nearest-K selection. Red dashed outlines show BL-BnB values.

For $K = 8$, BL-BnB raises the time-averaged per-snapshot maximum capacity from 520.6 to 523.8 Mbps (0.6%) and the minimum from 173.8 to 401.3 Mbps (Fig. 9). This suggests that better satellite choices and interference avoidance mainly benefit weaker links. Consistent with Fig. 8, the minimum-capacity gain falls from 227.5 to 34.4 Mbps as K increases from 8 to 49.

V. Conclusion

We studied intra-cell spatial reuse in LEO downlinks and developed BL-BnB to jointly account for link strength and receive-array leakage. The method combines branch-and-bound with an admissible upper bound and greedy-plus-swap initialization to obtain globally optimal schedules for the reference-point model. On the 24-h Starlink trace, BL-BnB improves mean per-beam capacity over nearest-satellite selection by 27.2% with eight beams and 9.8% with 49 beams, primarily benefiting weaker links. Increasing the number of overlapping beams raises total capacity sublinearly, while performance varies little across the studied cell radii. For selecting eight of 28 candidates, BL-BnB reduces mean runtime from 529.48 to 1.98 ms, although worst-case complexity remains exponential. These results support jointly considering link quality and interference for satellite spatial reuse.

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